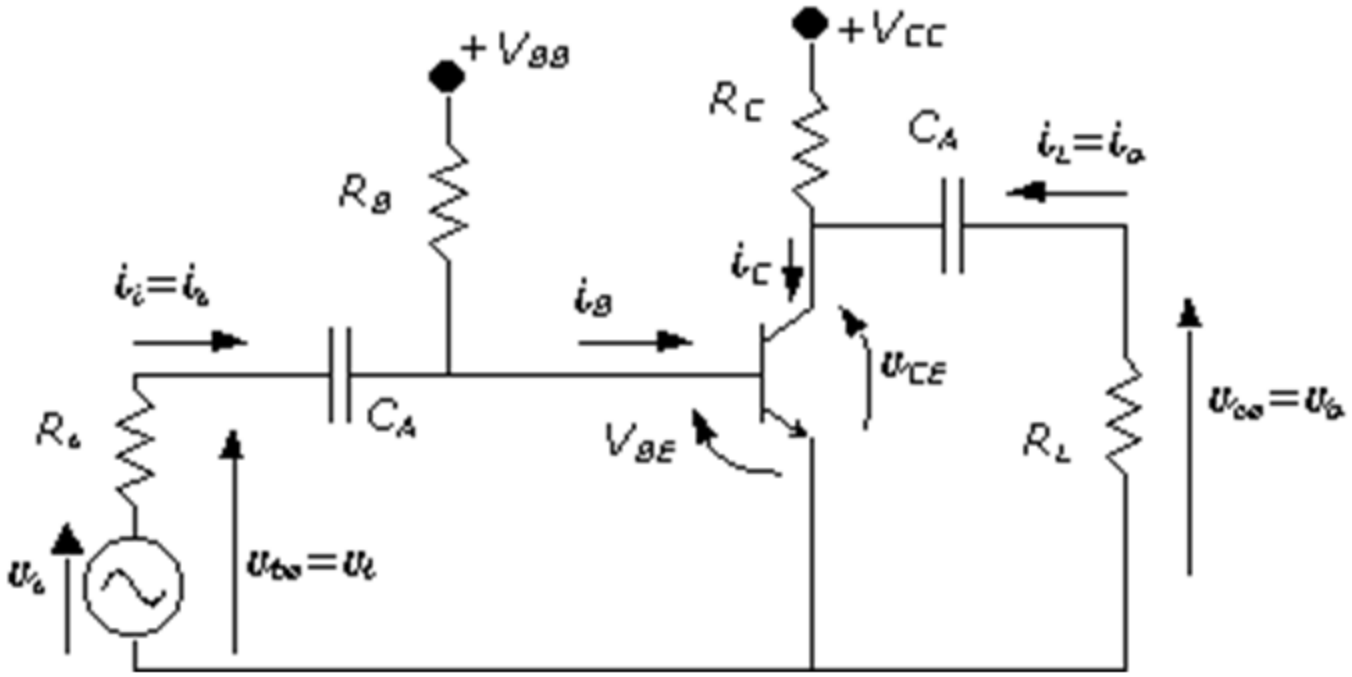
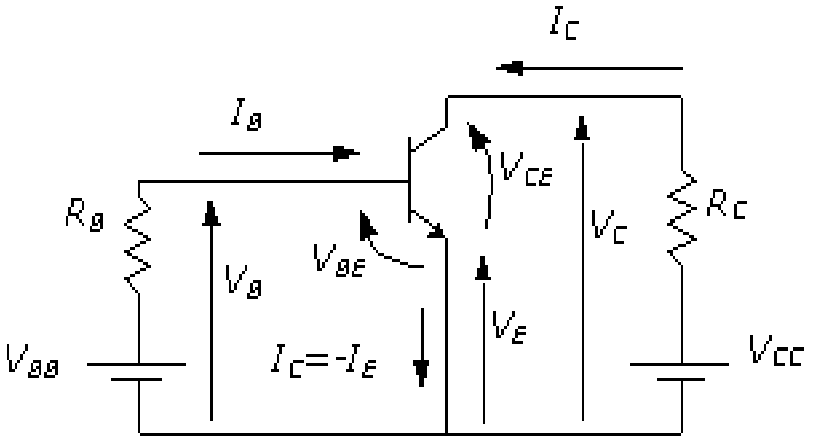
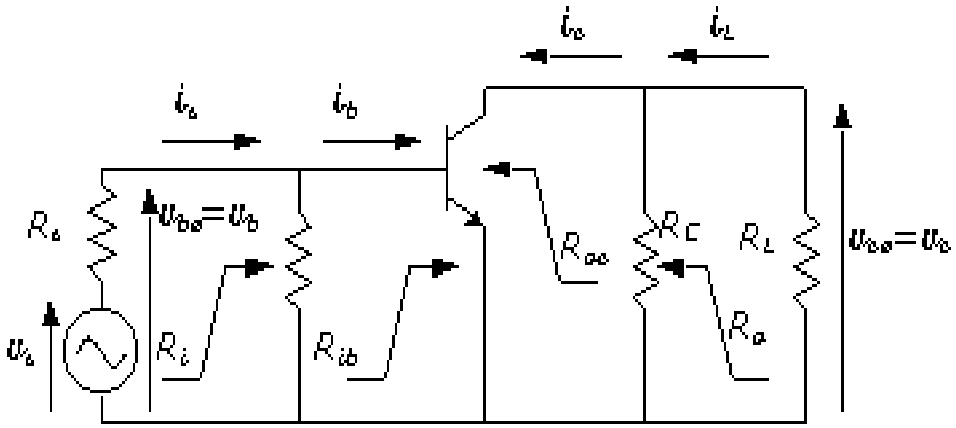
Amplificador básico







Continua o reposo (Q) ⟹ Señal a **frecuencias medias**

$$I_B = \frac{V_{BB} - V_{BE}}{R_B} = I_{BQ}$$ malla de ent.

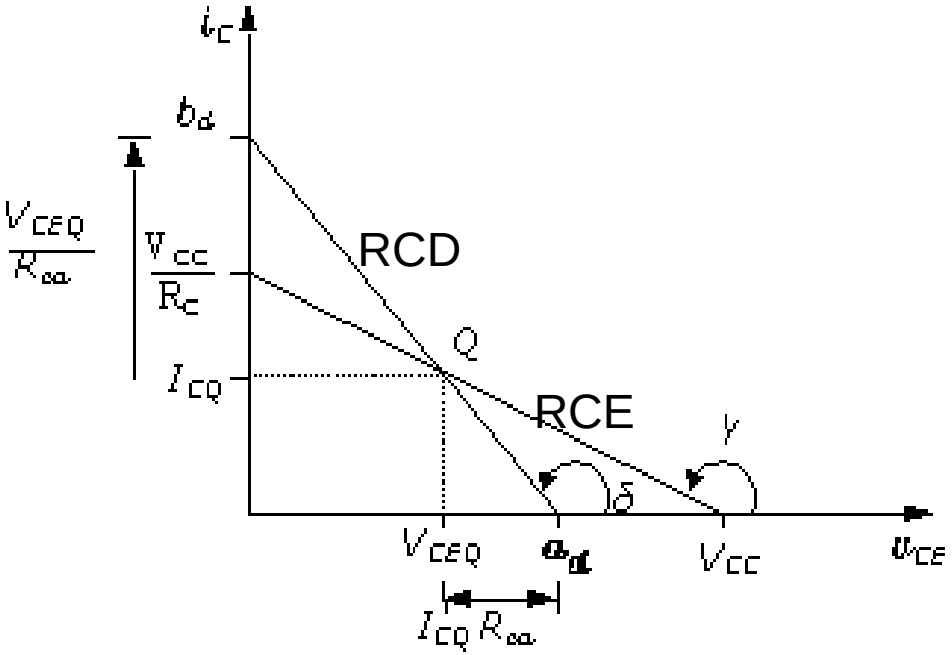
$I_{CQ} = \beta_F * I_{BQ}$ válida en MAD

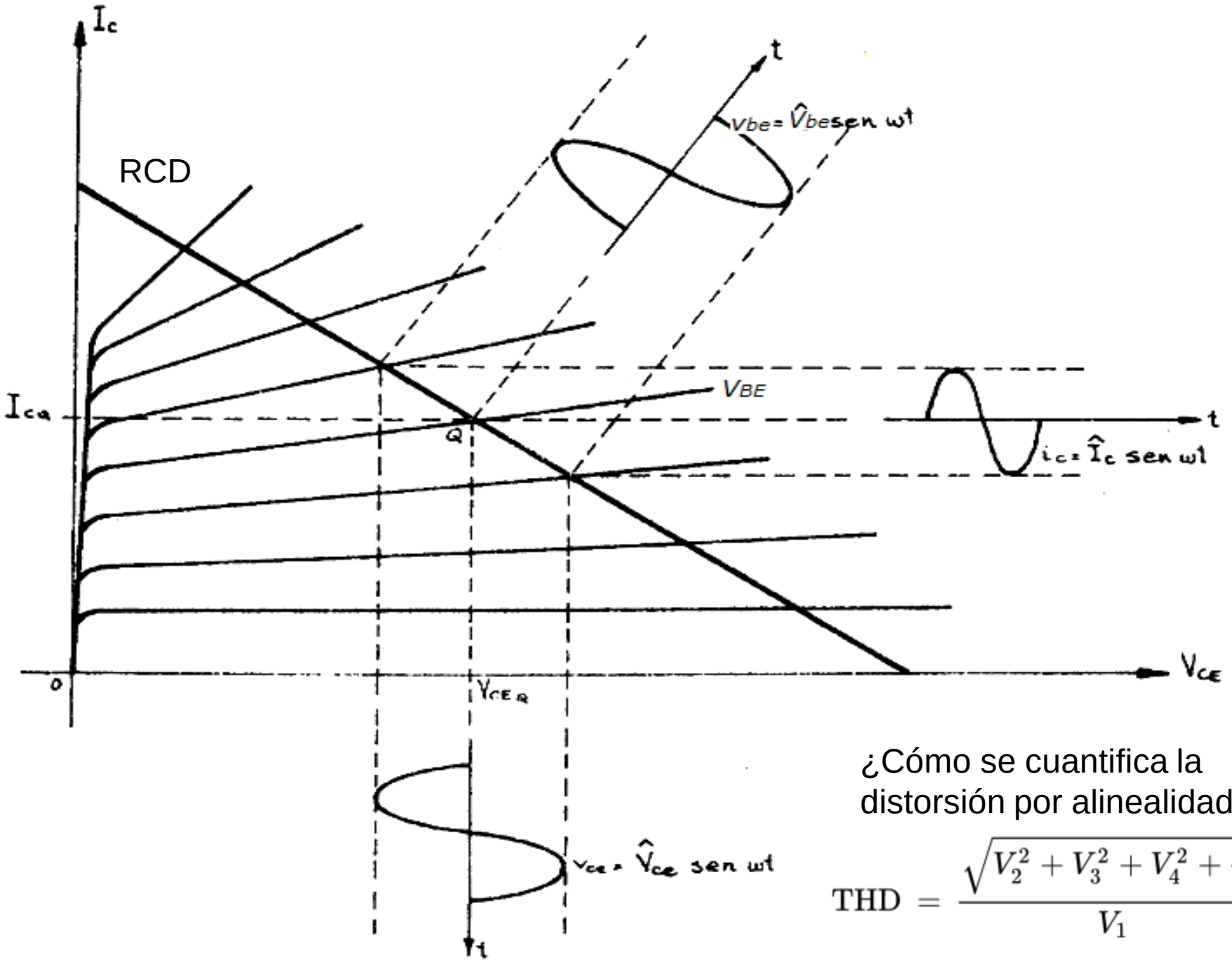
$V_{CEQ} = V_{CC} - I_{CQ} R_C$ malla de salida

V_{CEQ} **> 0**

¿Y si no es > 0? Se debe partir de otro modo de funcionamiento.

Con origen en Q: $\Delta v_{CE} = -\Delta i_C . R_{ca}$

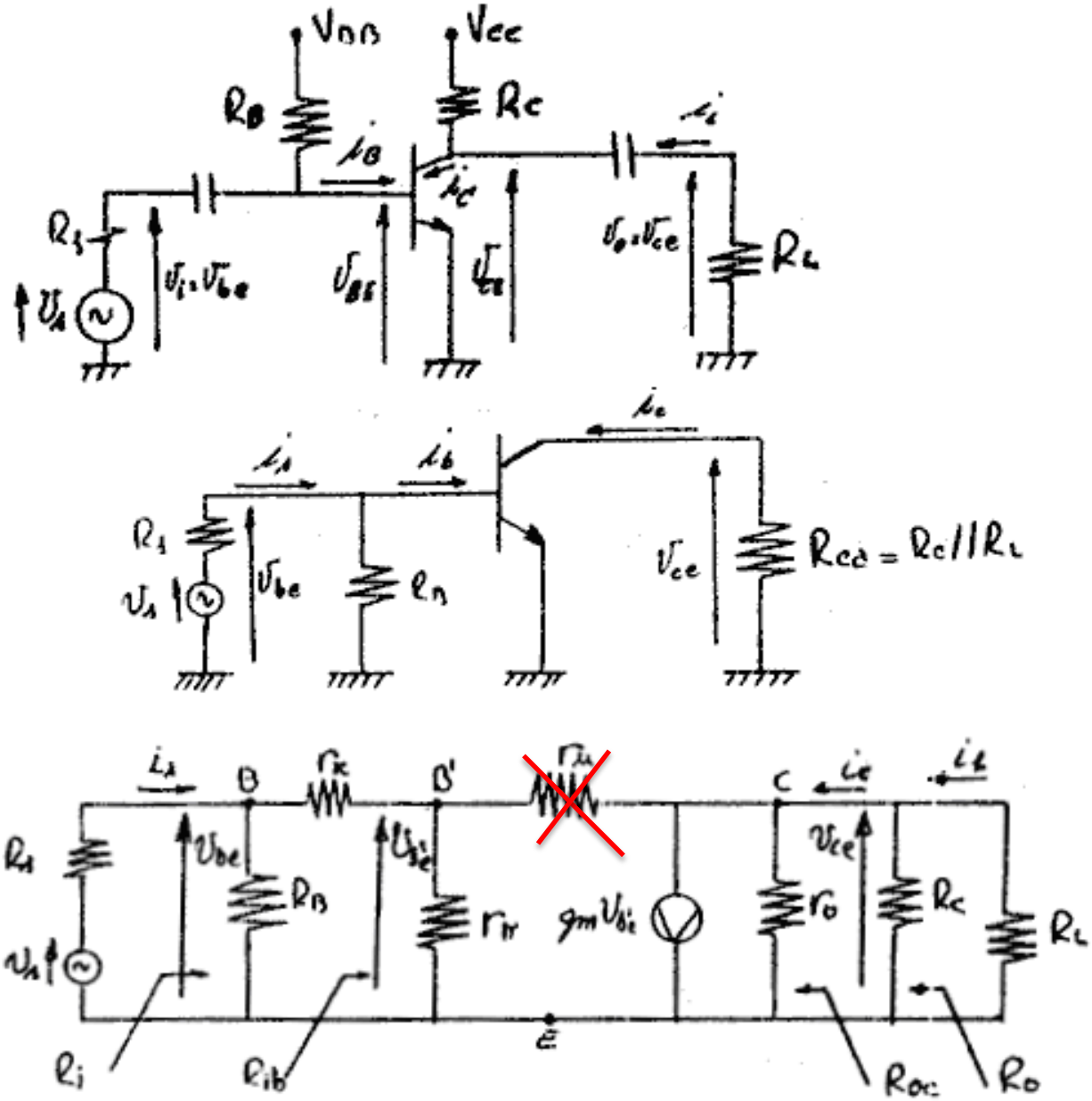


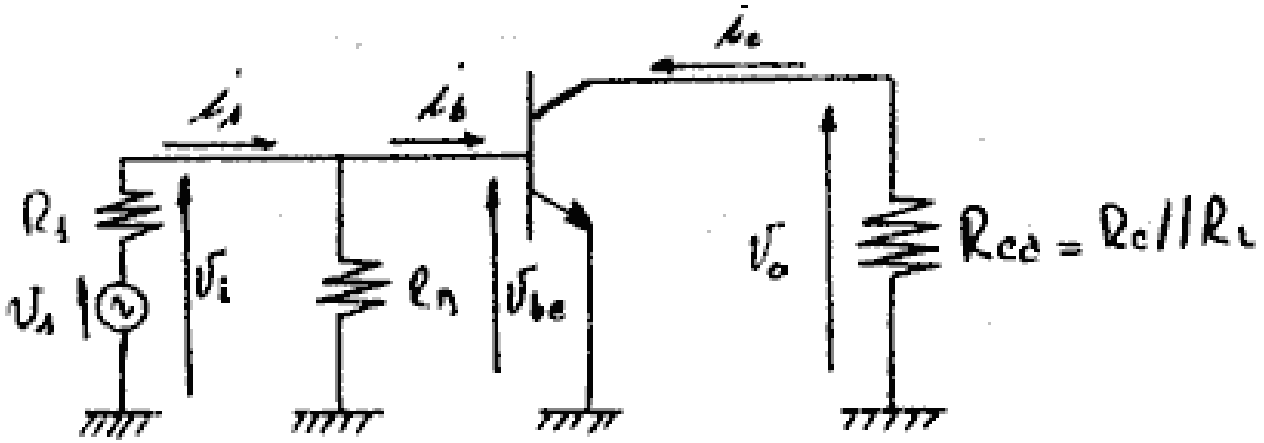


¿Cómo se cuantifica la distorsión por alinealidad?

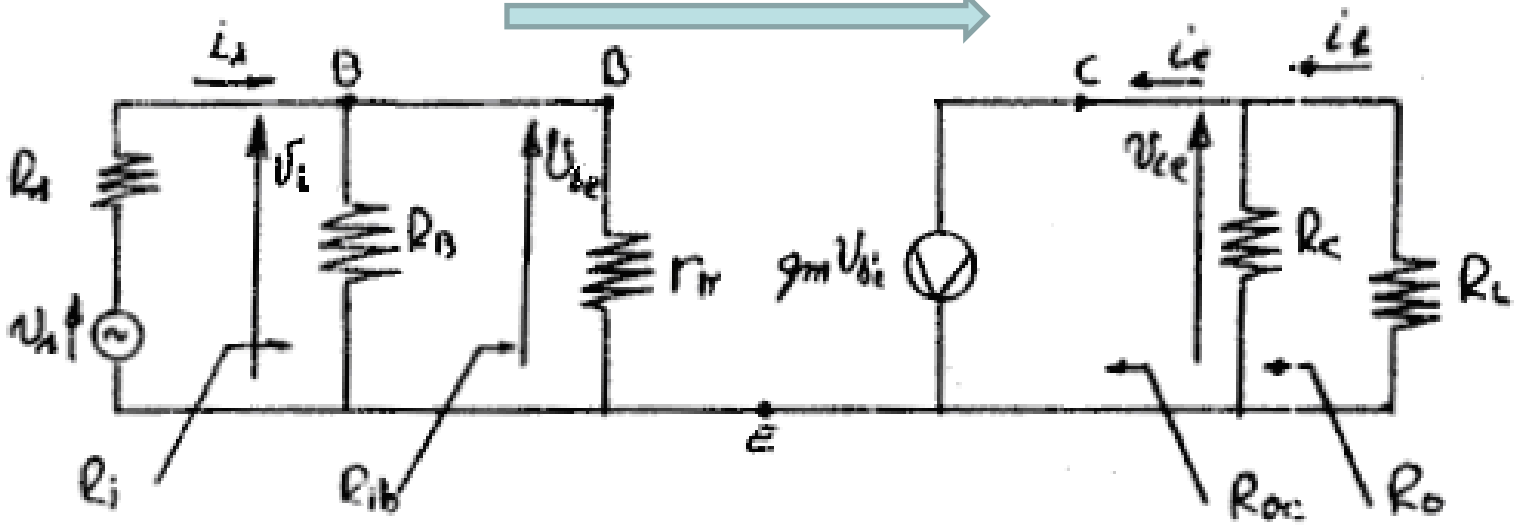
$$\mathrm{THD} = \frac{\sqrt{V_2^2 + V_3^2 + V_4^2 + \cdots}}{V_1}$$

Análisis en pequeña señal a frecuencias medias





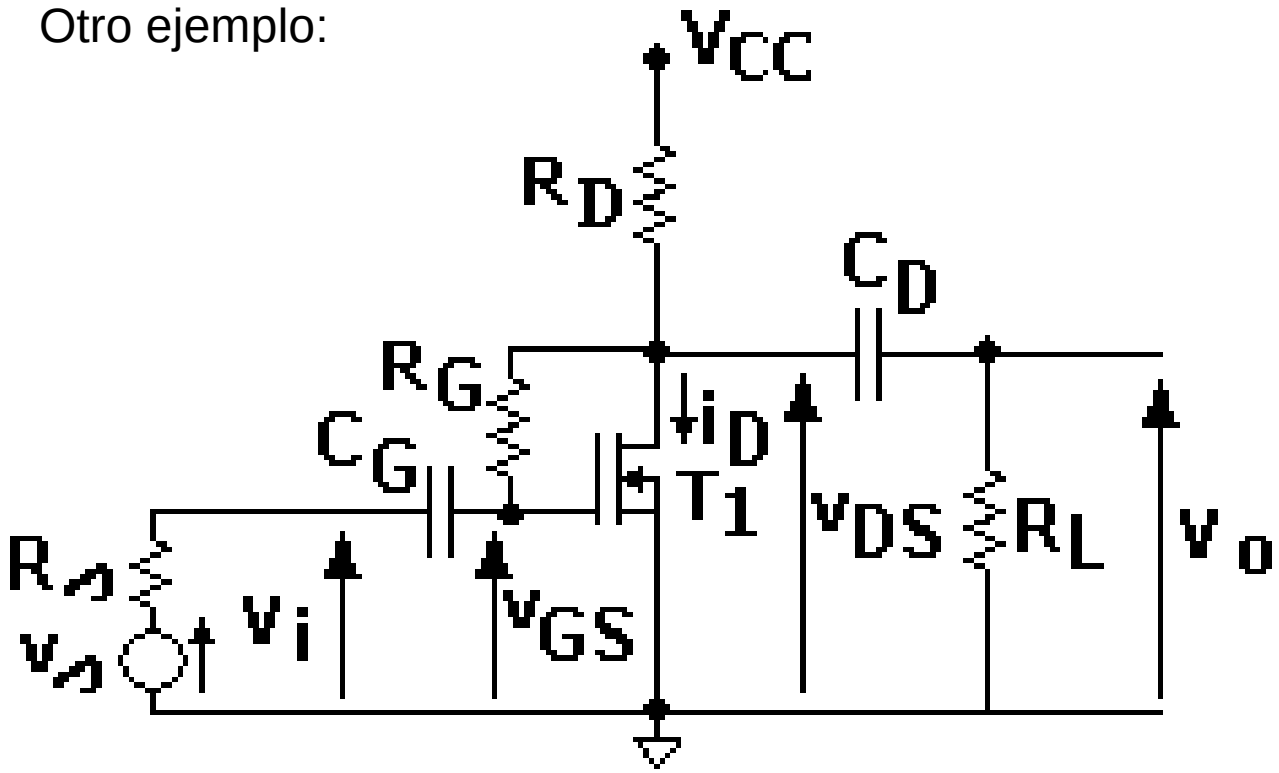
modelo simplificado (unilateral)



Análisis **por inspección:** aplicar herramientas de reducción sin usar explícitamente el modelo.

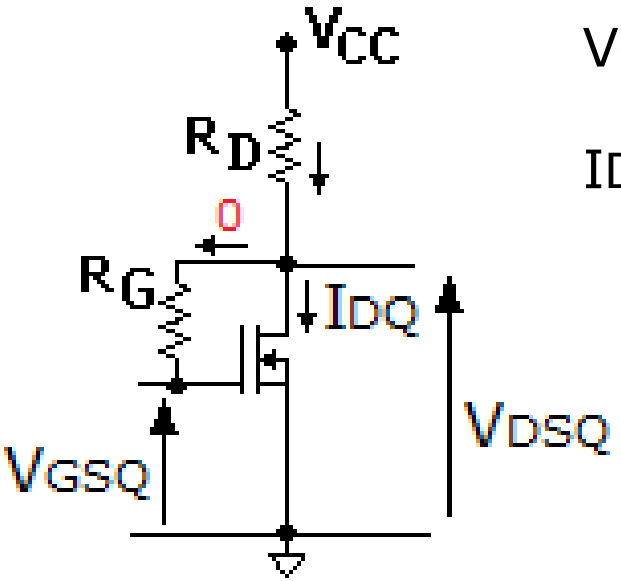
$$\frac{vo}{vi} = -\frac{ic\ Rca}{vbe} = -\ gm\ Rca = Av$$

Otro ejemplo:



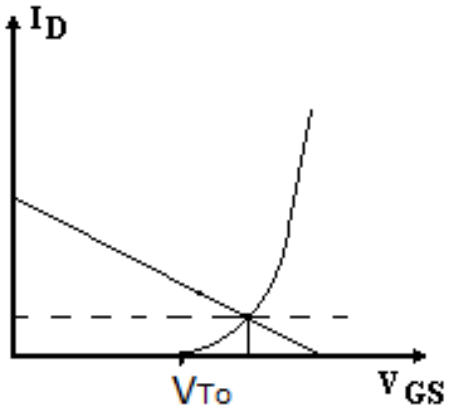
VCC = +24 V
RD = 5 KΩ
RG = 1 MΩ
Rs = 1 KΩ
RL = 50 KΩ
VTo = 3 V
K = 4 mA/V^2

Circuito de continua (Q):



$$V_{CC} - I_{DQ} R_D - V_{GSQ} = 0$$

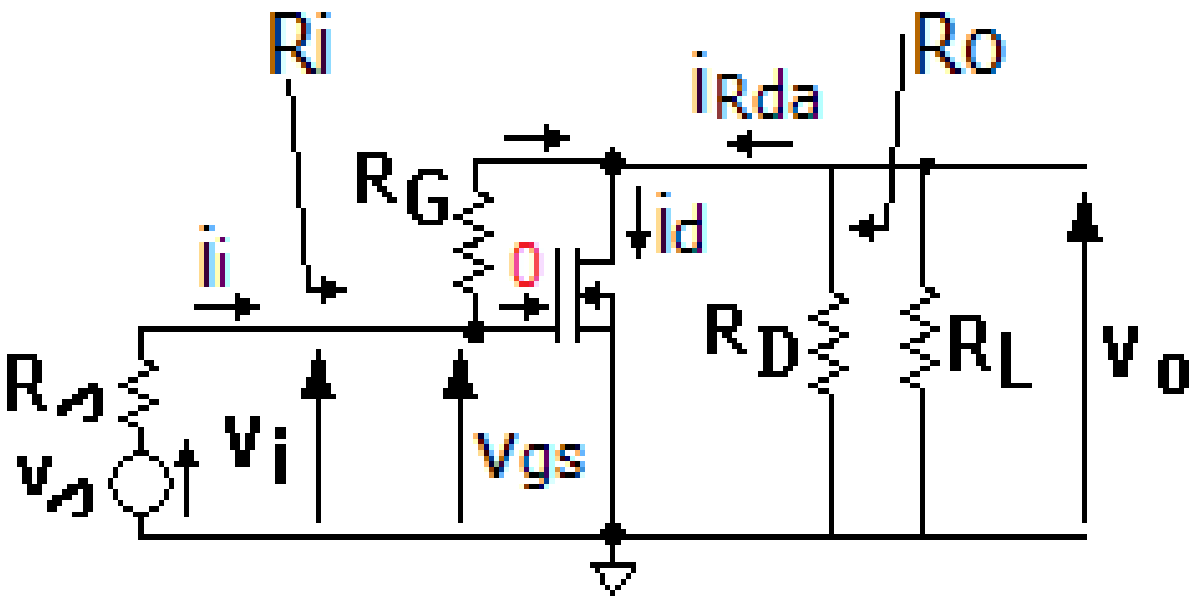
$$I_{DQ} = K (V_{GSQ} - V_{To})^2$$



$V_{GSQ} = 4\ V$; $I_{DQ} = 4\ mA$

$V_{DSQ} = 4\ V > V_{DSE} = 1V \rightarrow$ zona de control de potencia

Circuito de señal:



$i_{Rda} = - vo / (RD // RL)$

$ii = (vi - vo) / RG$

Si: $|Av| >> 1 \rightarrow vo >> vi$

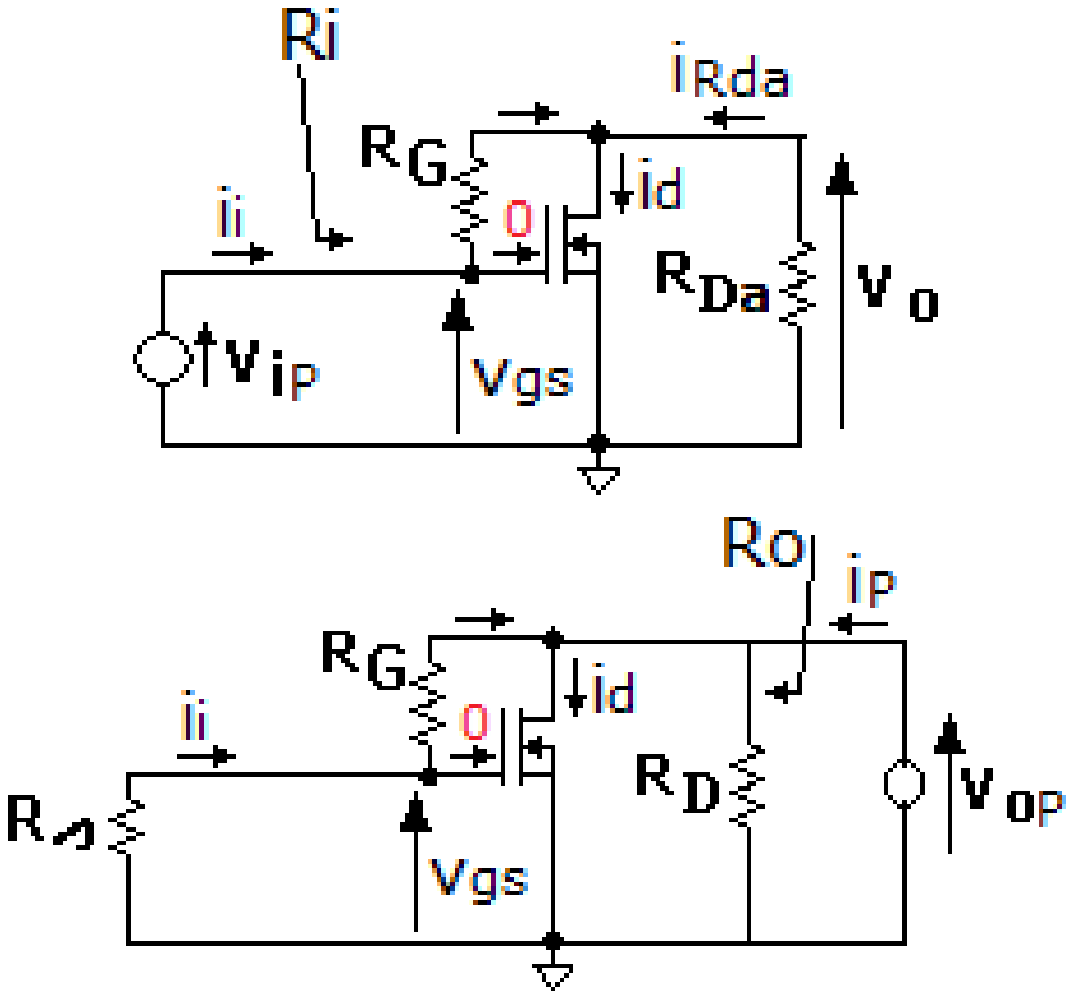
Entonces: $i_{Rda} \cong id$

$vo \cong - id\ (RD//RL) = - gm\ vgs\ (RD//RL) \rightarrow Av = vo / vi = - gm\ (RD//RL)$

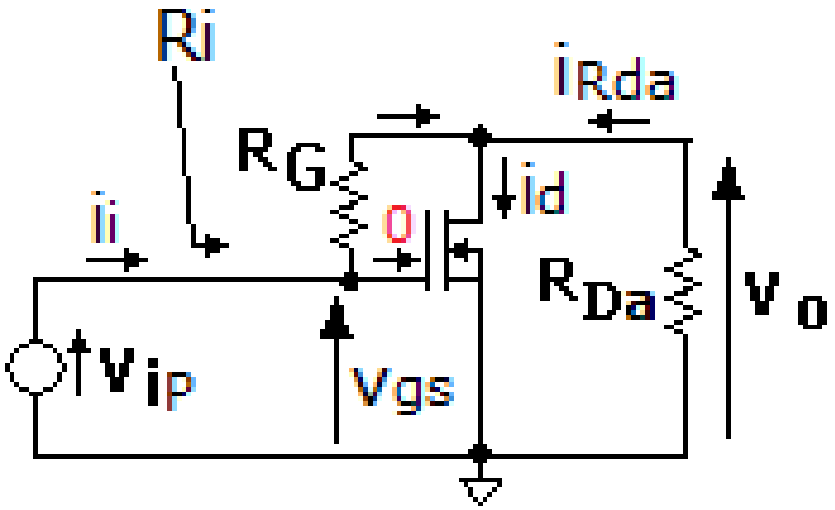
$= - 2\ mA/V\ (5\ k\Omega // 50\ k\Omega) \cong -10$

Bajo estas condiciones, *RG* no influye en Av

¿Cómo influye RG en Ri y Ro?:



Veamos Ri:

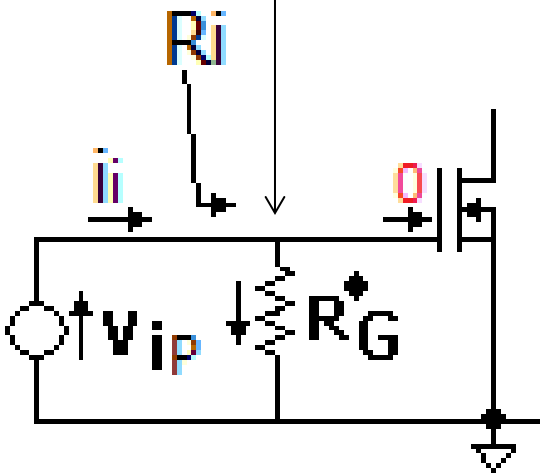


$ii = (vi - vo) / RG = (vi - Av\ vi) / RG \cong \dfrac{vi}{(RG / |Av|)}$

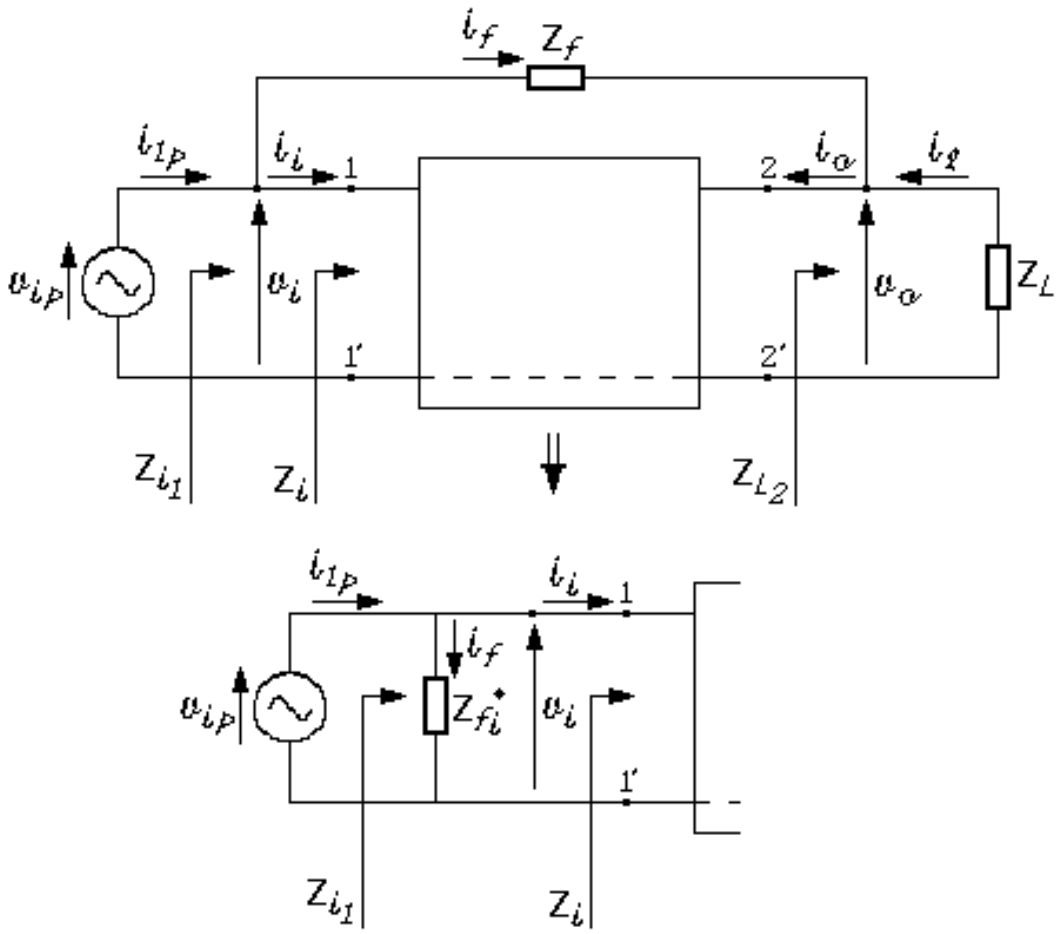
Sin simplificaciones:
$RG^* = RG / (1-Av)$

$\rightarrow Ri \cong 100\ K\Omega$

Reflexión por relación de V



Generalicemos:



Con |Av| >> 1 y Av < 0

Si Zf = R $\rightarrow$ *Zf* ≅ R / |Av|*

Si Zf $\rightarrow$ *L* $\rightarrow$ *Zf* ≅ jωL / |Av|*

Si Zf $\rightarrow$ *C* $\rightarrow$ *Zf* ≅ 1 / jωC |Av|*

Con |Av| >> 1 y Av > 0

Si Zf = R $\rightarrow$ *Zf* ≅ - R / |Av|*

¿es posible?

Con Av ≅ 1

*Zf** $\rightarrow \infty$

Recordar que sin simplificaciones:

Zf = Zf / (1-Av)*